



ECE3441_SP
25_Quiz4B...

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ECE 3441 Spring 2025 Quiz 4

Version B

- Closed book, closed notes
- No cheat sheet is allowed.
- Only four function calculators are allowed.
- Show all details of your work to obtain credit (The answer alone will not suffice).
- You may not talk during the quiz to others or to yourself.
- You have 45 minutes.

Please do not open booklet unless instructed to do so, or substantial penalty will result.

SOLUTIONS

Last Name, First Name Peoplesoft ID, Date (YYYY/MM/DD)

Question	Points	Maximum
1		25
2		25
3		30
4		30

Total: _____/(100 possible points + 10 bonus points)

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Question #1:

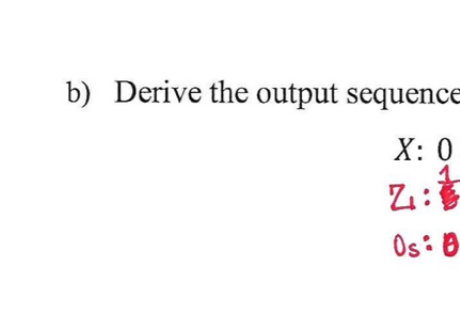
Consider a sequential circuit which has one input (X), one output (Z), and the following behavior.

$Z = 1$ if the total number of 0s received is odd OR the two most recent inputs have been 1.

Note: zero is an even number; the OR indicates that either condition is enough for $Z = 1$.

- a) Draw a Mealy state graph for the circuit. Hint: the optimal number of states is three.

States: *Does not rely on previous input since condition is already met*
 S_0 : odd 0s, 0 received
default: S_1 : even 0s, 0 received
 S_2 : even 0s, 1 received
** Draw what meets conditions first*



- b) Derive the output sequence for the following input sequence.

X : 0 0 1 1 1 0 0 1 1 0 0 0 0 1 1 1 1 1
 Z : 1 0 0 1 1 1 0 0 1 1 0 0 1 1 1 1 1
 $0s$: 0 E E E E 0 E E 0 E 0 E E 0 0 0 0 0 0

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Question #2:

Design a Moore machine with one input (X), two outputs (Z_1 and Z_2), and the following behavior.

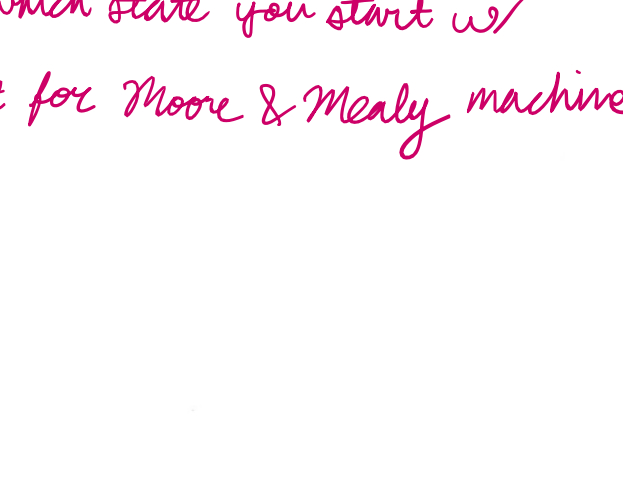
$Z_1 = 1$ if the total number of 0s which the machine has received is even, and $Z_1 = 0$ otherwise.

$Z_2 = 1$ if the total number of 1s which the machine has received is odd, and $Z_2 = 0$ otherwise.

Hint: zero is an even number; the minimal number of states is four.

Note: providing a state graph is sufficient for full credit.

State	0s	1s	Z_1	Z_2
S_0	Even	Even	1	0
S_1	Odd	Even	0	0
S_2	Even	Odd	1	1
S_3	Odd	Odd	0	1



* Multiple answers depending on which state you start w/
* Know how to do input/output for Moore & Mealy machine (state graph)

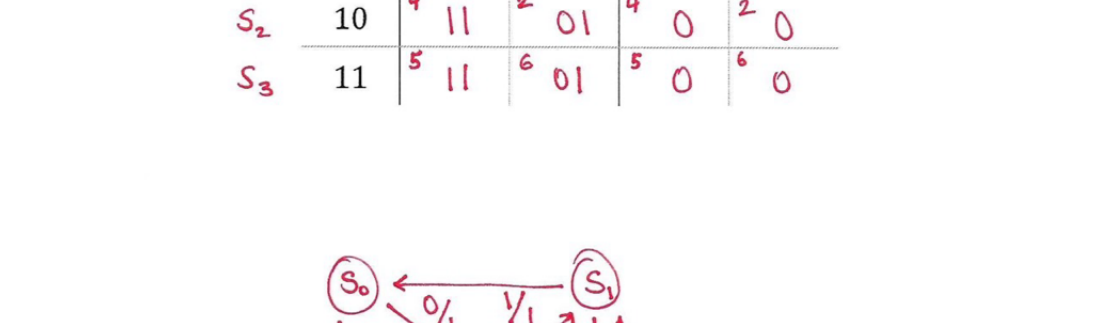
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Question #3:

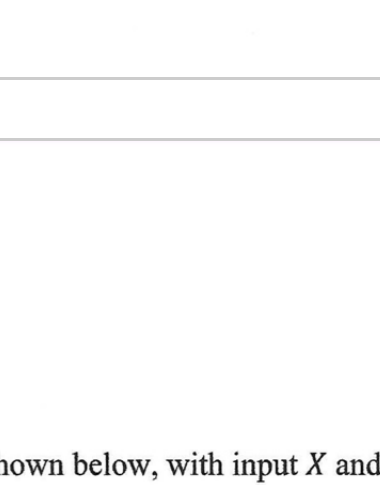
A Mealy sequential circuit has one input, one output, and two falling-edge-triggered flip-flops.

Construct a transition table and state graph for the circuit based on the following timing diagram.

Hint: for example, at the first falling edge, $Q_1Q_2 = 00$ and $X = 0$, then, $Q_1^+Q_2^+ = 10$ and $Z = 1$; similarly, look at the other falling edges and the variables to understand the circuit's behavior.



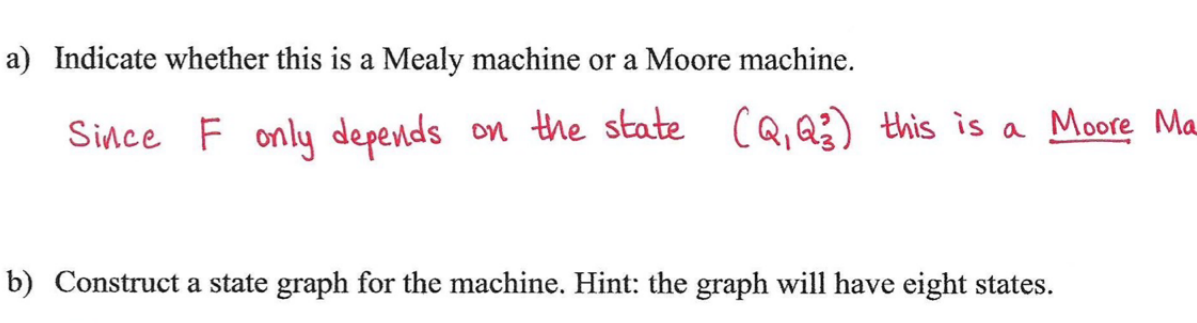
	Q_2Q_1	$X=0$	$X=1$	Z
S_0	00	1	0	1
S_1	01	0	1	1
S_2	10	1	0	0
S_3	11	0	1	0



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Question #4:

Consider the shift register shown below, with input X and output F .

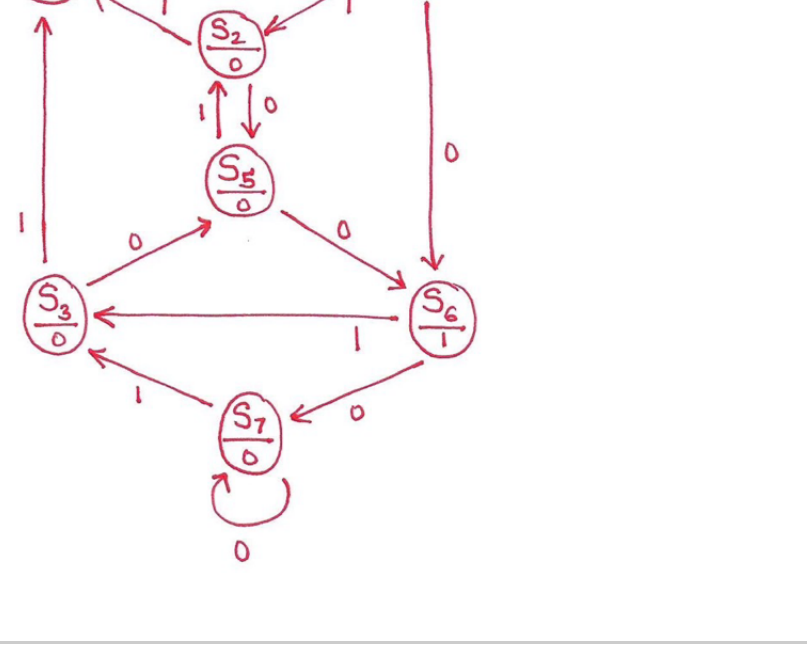


- a) Indicate whether this is a Mealy machine or a Moore machine.

Since F only depends on the state ($Q_1Q_2Q_3$) this is a Moore Machine.

- b) Construct a state graph for the machine. Hint: the graph will have eight states.

Next	$X=0$	$X=1$	F
S_0 : 000	100	000	0
S_1 : 001	101	000	0
S_2 : 010	100	001	0
S_3 : 011	101	001	0
S_4 : 100	110	010	1
S_5 : 101	110	010	0
S_6 : 110	111	011	1
S_7 : 111	111	011	0

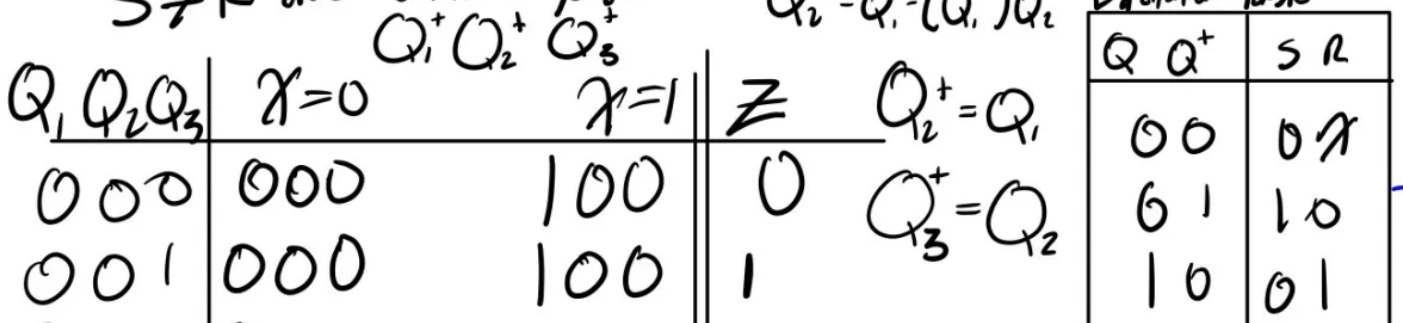


13.2-13.5, 13.13, 13.15, 13.21, 13.27, 13.30

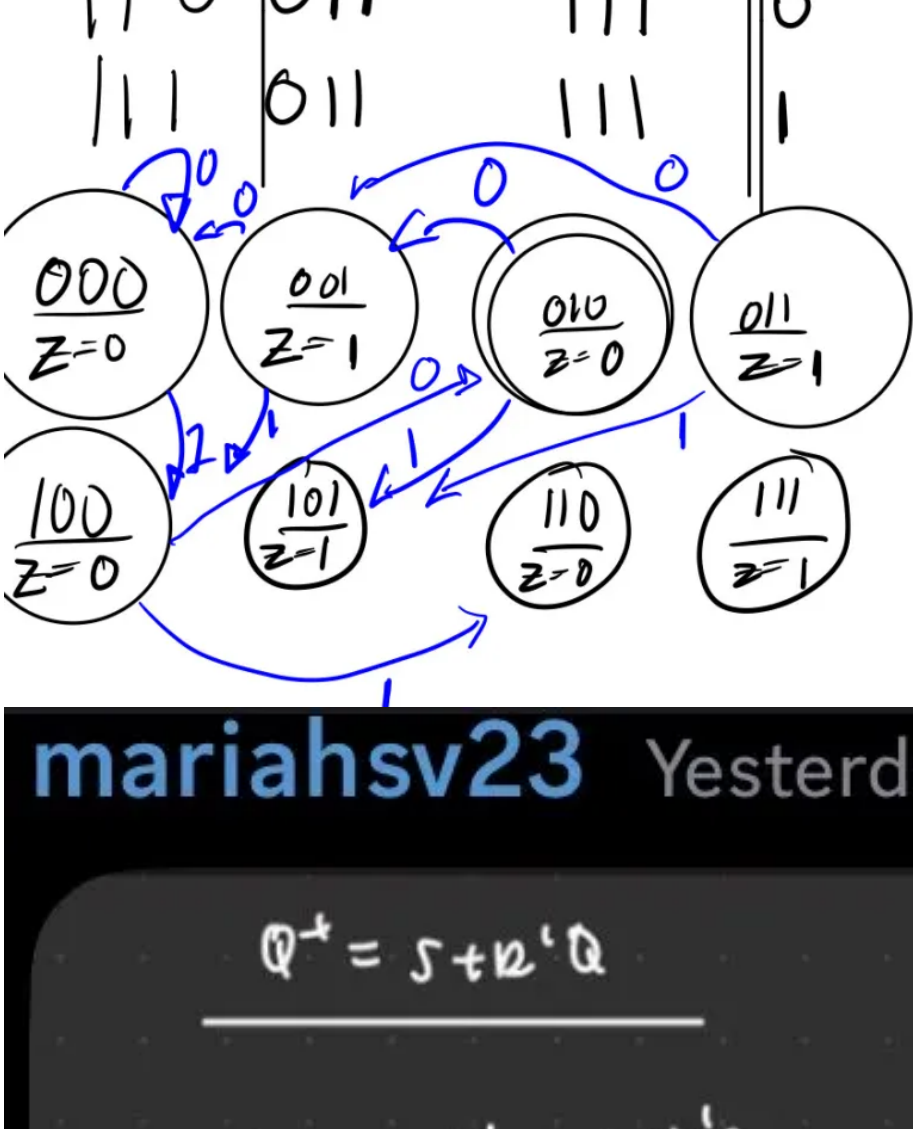
13.2 Construct a transition graph for the shift register shown. (X is the input, and

Z is the output.) Is this a Mealy or Moore machine?

$Z = Q_3^+$ SR T.T. $Q_1^+ = S + RQ_1$, $Q_2^+ = Q_1$, $Q_3^+ = Q_2$



$Q_1Q_2Q_3$	$X=0$	$X=1$	Z
000	000	100	0
001	000	100	0
010	001	101	0
011	001	101	0
100	010	110	0
101	010	110	0
110	011	111	0
111	011	111	0



$Q_1^+ = S + RQ_1$
 $Q_2^+ = Q_1$
 $Q_3^+ = Q_2$

$Q_1^+ = S + RQ_1$
 $Q_2^+ = Q_1$
 $Q_3^+ = Q_2$

$Q_1^+ = S + RQ_1$
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